

Hotel Billing & Restaurant Management System

1. Project Overview

The Hotel Billing & Restaurant Management System is a cloud-based application designed to streamline restaurant operations by connecting waiters, chefs, and managers through a centralized platform.

The system consists of three applications:

1. Waiter Mobile Application
2. Chef/Kitchen Application
3. Manager Web Dashboard

All applications communicate with a centralized backend hosted on the cloud, allowing users to access real-time information from anywhere.

2. Project Objectives

- Digitize restaurant order management.
 - Eliminate manual order slips.
 - Provide real-time communication between waiters and chefs.
 - Generate accurate bills automatically.
 - Track sales and restaurant performance.
 - Maintain centralized data accessible from any location.
 - Demonstrate cloud deployment, DevOps, and modern software architecture.
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3. User Roles

3.1 Admin

Responsibilities:

- Manage users
- Manage roles and permissions
- View system analytics
- Configure restaurant settings

3.2 Manager

Responsibilities:

- View dashboard
- Manage menu items
- Generate bills

- View reports
- Monitor orders
- Manage staff

3.3 Waiter

Responsibilities:

- Login to system
- View tables
- Create orders
- Update orders
- Send orders to kitchen
- Track order status

3.4 Chef

Responsibilities:

- View incoming orders
- Update order status
- Mark orders as preparing
- Mark orders as ready
- View completed orders

4. System Architecture

Waiter Mobile App | | V Spring Boot Backend API | | PostgreSQL Database |

| | V V Chef App Manager Dashboard

All applications communicate through REST APIs and WebSockets.

5. Technology Stack

Frontend

Waiter Application

- Android Studio
- Java
- Material Design

Chef Application

- Android Studio
- Java
- Material Design

Manager Dashboard

- React
 - TypeScript
 - Tailwind CSS
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Backend

- Java 21
 - Spring Boot
 - Spring Security
 - Spring Data JPA
 - Spring WebSocket
 - JWT Authentication
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Database

- PostgreSQL
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Cloud Services

AWS Services:

- EC2
 - RDS PostgreSQL
 - S3
 - CloudWatch
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DevOps

- Git
 - GitHub
 - Docker
 - Azure DevOps Pipelines
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6. Core Features

Authentication Module

Features:

- Login
- Logout
- JWT Authentication
- Role-Based Access Control

Roles:

- Admin
 - Manager
 - Waiter
 - Chef
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Menu Management Module

Features:

- Add Menu Item
 - Update Menu Item
 - Delete Menu Item
 - View Menu Categories
 - Upload Food Images
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Order Management Module

Features:

- Create Order
- Update Order
- Cancel Order
- View Order Status

Order Status:

- Pending
 - Preparing
 - Ready
 - Completed
 - Cancelled
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Kitchen Management Module

Features:

- Receive Orders Instantly
 - Update Preparation Status
 - Mark Orders Ready
 - View Completed Orders
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Billing Module

Features:

- Generate Bill
 - Apply Tax
 - Apply Discount
 - Print Receipt
 - Download PDF Invoice
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Reporting Module

Features:

- Daily Sales Report
 - Weekly Sales Report
 - Monthly Sales Report
 - Most Sold Items
 - Revenue Analytics
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Inventory Module

Features:

- Manage Ingredients
 - Track Stock Levels
 - Low Stock Alerts
 - Inventory Reports
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7. Database Design

Users Table

Fields:

- user_id
- name
- email
- password
- role
- created_at

Tables Table

Fields:

- table_id
- table_number
- capacity
- status

Menu Items Table

Fields:

- item_id
- item_name
- category
- price
- image_url

Orders Table

Fields:

- order_id
- table_id
- waiter_id
- status
- created_at

Order Items Table

Fields:

- order_item_id
- order_id
- item_id

- quantity
- price

Bills Table

Fields:

- bill_id
 - order_id
 - subtotal
 - tax
 - discount
 - total_amount
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8. Real-Time Communication

The system will use Spring WebSocket to enable instant communication.

Workflow:

1. Waiter places an order.
 2. Backend receives the order.
 3. WebSocket sends the order to the Chef Application.
 4. Chef updates order status.
 5. Waiter receives status updates instantly.
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9. AWS Infrastructure

Amazon EC2

Purpose:

- Host Spring Boot Backend

Amazon RDS

Purpose:

- PostgreSQL Database Hosting

Amazon S3

Purpose:

- Store Bills

- Store Reports
- Store Food Images

Amazon CloudWatch

Purpose:

- Application Monitoring
 - Log Management
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10. CI/CD Pipeline

Workflow:

Developer Pushes Code | V GitHub Repository | V Azure DevOps Pipeline | V Build Application | V Run Automated Tests | V Create Docker Image | V Deploy to AWS EC2

Benefits:

- Faster deployments
 - Automated testing
 - Consistent releases
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11. Future Enhancements

- QR-Based Table Ordering
 - Online Payment Integration
 - Customer Feedback System
 - AI-Based Sales Prediction
 - Multi-Branch Restaurant Support
 - Loyalty Program
 - SMS Notifications
 - Mobile Dashboard for Managers
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12. Development Roadmap

Week 1

- Requirements Analysis
- Database Design
- ER Diagram
- API Planning

Week 2

- Authentication Module
- User Management Module

Week 3

- Menu Management Module

Week 4

- Order Management Module

Week 5

- Kitchen Management Module
- WebSocket Integration

Week 6

- Billing Module

Week 7

- Reporting Module

Week 8

- AWS Deployment
- Azure DevOps Pipeline Setup
- Final Testing

Expected Outcome

A production-ready cloud-based restaurant management system capable of handling restaurant operations through mobile and web applications while demonstrating expertise in Java, Spring Boot, PostgreSQL, AWS Cloud Services, Android Development, React, and DevOps practices.